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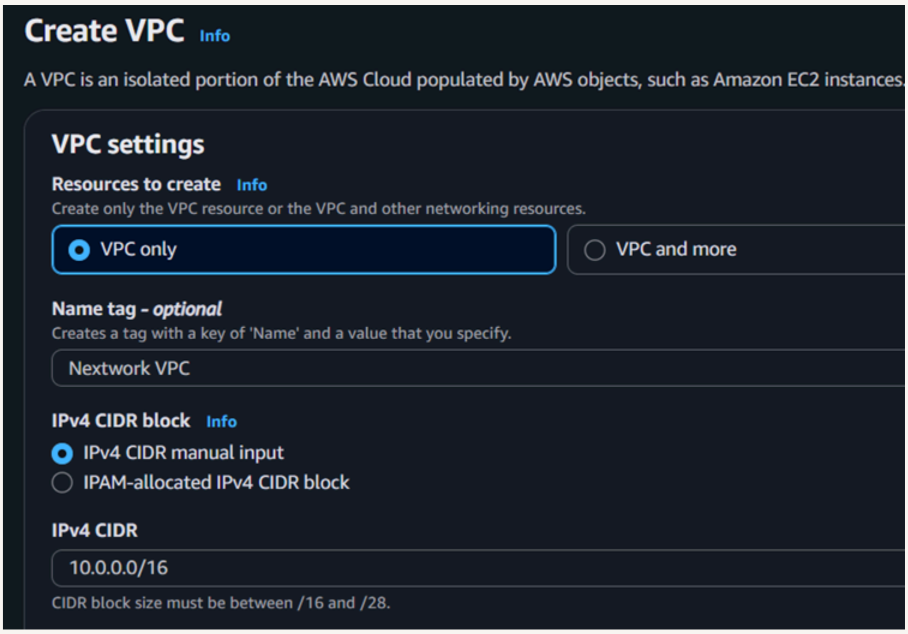
Build a Virtual Private Cloud (VPC)

Amir Hamzah



Introducing Today's Project!

What is Amazon VPC?

Amazon VPC is a virtual network dedicated to your AWS account, and it is useful because it allows you to securely control and customize your cloud networking environment.

How I used Amazon VPC in this project

In today's project, I used Amazon VPC to create a secure and customizable virtual network where I launched a public subnet and attached an internet gateway to enable internet access for resources within that subnet.

One thing I didn't expect in this project was...

One thing I didn't expect in this project was that even after creating the VPC and internet gateway, additional configurations were still needed to make everything work.

This project took me...

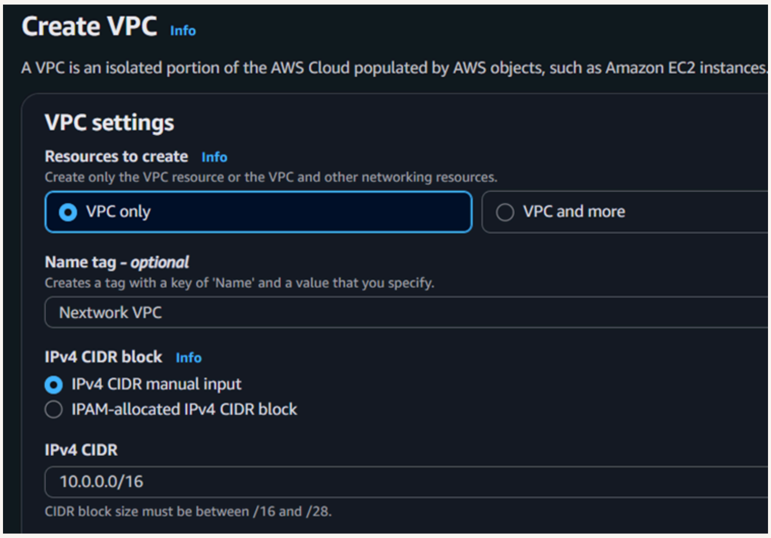
This project took me a little over an hour to complete. Most of the time was spent understanding how the different components like the VPC, public subnet, and internet gateway.

Virtual Private Clouds (VPCs)

VPCs are logically isolated section of the AWS cloud where you can launch AWS resources within a virtual network you define. Think of it as a private space within AWS where you can control your network configuration.

There was already a default VPC in my account ever since my AWS account was created. It is to simplify the initial setup and allow users to quickly launch resources without needing to configure a custom VPC.

To set up my VPC, I had to define an IPv4 CIDR block, which is means a range of IP addresses that my VPC can allocate to the resources deployed into my VPC.

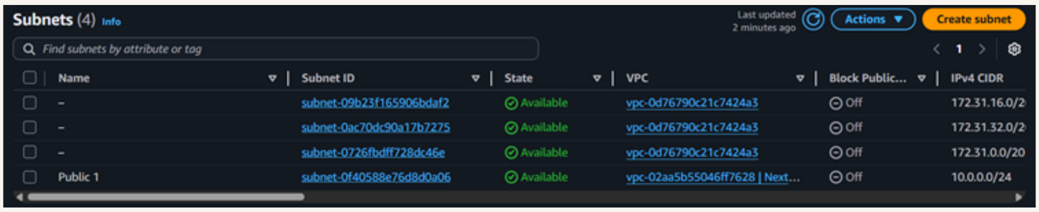

Create VPC Info
A VPC is an isolated portion of the AWS Cloud populated by AWS objects, such as Amazon EC2 instances.
VPC settings
Resources to create Info
Create only the VPC resource or the VPC and other networking resources.
VPC only
VPC and more
Name tag - optional
Creates a tag with a key of 'Name' and a value that you specify.
Nextwork VPC
IPv4 CIDR block Info
IPv4 CIDR manual input
IPAM-allocated IPv4 CIDR block
IPv4 CIDR
10.0.0.0/16
CIDR block size must be between /16 and /28.

Subnets

Subnets are subsections of my VPC,just like how neighbourhoods are subsections of a city. There are already subnets existing in my account, one for every Availability Zones in the Region that I've set up my VPC in.

Once I created my subnet, I enabled auto-assign public IPV4 address.This setting makes sure the EC2 instances get the public adress automatically so that I do need to do it manually.

The difference between public and private subnets in AWS is that, for a subnet to be considered public, it must have a route to the internet through an internet gateway.

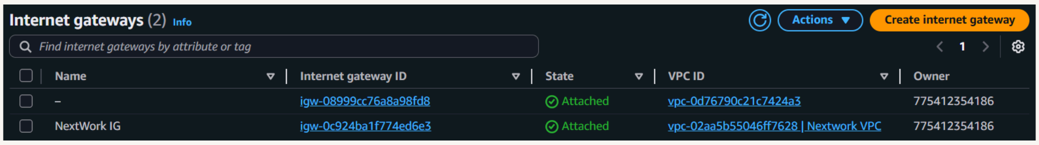


Name	Subnet ID	State	VPC	Block Public...	IPv4 CIDR
–	subnet-09b23f165906bdaf2	Available	vpc-0d76790c21c7424a3	Off	172.31.16.0/2
–	subnet-0ac70dc90a17b7275	Available	vpc-0d76790c21c7424a3	Off	172.31.32.0/2
–	subnet-0726fbdff728dc46e	Available	vpc-0d76790c21c7424a3	Off	172.31.0.0/20
Public 1	subnet-0f40588e76d8d0a06	Available	vpc-02aa5b55046ff7628 | Next...	Off	10.0.0.0/24

Internet gateways

Internet gateways are the key VPC components that allows internet access for the resources in my VPC/subnet.An internet gateway is also how users in the public can access my resources in my public subnet.

Attaching an internet gateway to a VPC means resources in your VPC can now access the internet.The EC2 instances with public IP addresses also become accessible to users, so your applications hosted on those servers become public too.



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